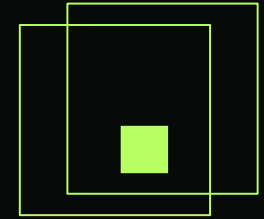


# CR3DX



## Credit history that follows the borrower across chains

Money moves on Ethereum. Credit state lives on Creditcoin.  
Attestcoin is the only connection, and anyone can submit the proof.

---

SOLO BUILDER

**Valery Borovsky**

[Repository »](#) [Live dashboard »](#)

# Repayment history stops at the chain boundary

**A borrower can repay reliably on one chain and still arrive on the next with no portable record.**

The history exists.  
It just cannot be proven where the next lender needs it.

SOURCE CHAIN

**RECORD**

of on-time repayments

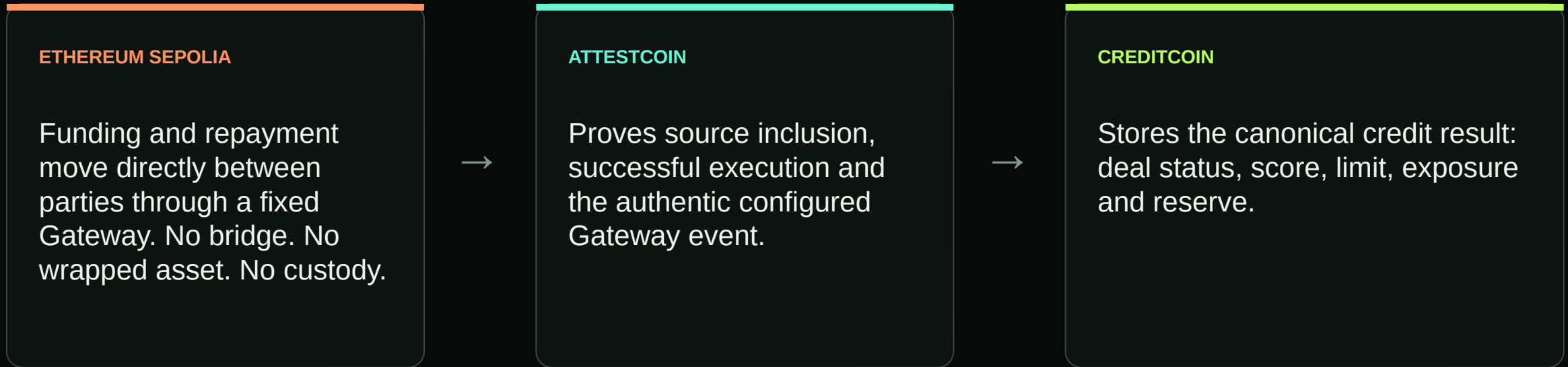
NEXT CHAIN

?

no verifiable history

This matters in cross-border invoice financing, where payer, borrower and lender often use different rails.

# Settlement stays with the money. Credit state stays with the history.



**Without Attestcoin, Cr3dX cannot verify source funding or repayment facts or evaluate source-chain deadlines.**

The proof is the boundary. The credit state is the result.

# A repayment arrived first; the system still closed correctly

CANONICAL OUTCOME

**PAID\_ON\_TIME**

Repayment proof → VERIFIED\_PENDING

Funding proof → APPLIED

Targeted applyEvidence call → APPLIED

500 → 525

credit score

5,000 → 5,250

credit limit · USDC

1.0000001

funded · USDC

1.1

repaid on time · USDC

0

exposure / outstanding

Small testnet amounts by design; the limit demonstrates model behavior, not deal size.

Accepted snapshot: [f359c54](#) · dashboard separately re-reads current state from public RPC

REPAYMENT PROOF  
[a626556e...55657](#) »

FUNDING PROOF  
[c740cf0e...822ad](#) »

REPAYMENT APPLY  
[a0c24a10...b13b6](#) »

EXTERNAL RACE  
[8d859033...840c2](#) »

# We say exactly what the proof establishes, and what it does not

## PROVEN CRYPTOGRAPHICALLY

- ✓ Funding transaction inclusion, successful receipt and authentic Gateway event
- ✓ Repayment transaction inclusion, successful receipt and authentic Gateway event
- ✓ Deadline evaluation against attested source-chain height
- ✓ Credit outcome derived from those facts and contract rules

## NOT PROVEN, AND NOT CLAIMED

- ✗ That a real invoice exists or the claim is legally enforceable
- ✗ The borrower's real-world identity
- ✗ Sybil resistance: the score is address-level repayment reputation
- ✗ Collateral, escrow or collection enforcement

### TOKENS MOVED

This follows from the Gateway contract itself: it emits only after a successful transfer. The contract is small, public and has no privileged functions.

# The model and the implementation met only after expectations were sealed

# 63/63

traces matched  
zero semantic divergences

- |    |                              |                                                                |
|----|------------------------------|----------------------------------------------------------------|
| 01 | <b>Specification only</b>    | Isolated model workspace had no access to Deals or Credit code |
| 02 | <b>Expectations exported</b> | Trace corpus hashed before code access                         |
| 03 | <b>Implementation opened</b> | Foundry replayed the sealed oracle                             |
| 04 | <b>Convergence</b>           | All 63 traces matched                                          |

**16 independently derived**

strongest evidence

**47 specification-prescribed**

full coverage, but less independent

[Open report and manifests »](#)

Not a formal proof of every reachable state. An independently built model and Solidity implementation agreed on the entire pre-sealed corpus.

# The external race ended with zero worker signatures

# 0

worker signatures

0 broadcasts · nonce unchanged

**A second wallet submitted the proof first.**

The worker read the on-chain state, found the fact already recorded and closed the task without signing or broadcasting. The deal state remained correct.

#### WHY THIS MATTERS

The worker can deliver valid facts, but cannot forge one or force a transition the public contracts do not permit.

**No worker-only contract method exists. A test-derived allowlist fails if a third state-changing worker method appears.**

[Inspect the S6 coverage matrix »](#)

# Building on Attestcoin meant measuring its operational boundary

## GAS ACCOUNTING

`verify` and `verifyAndEmit` used identical gas on identical inputs. The team indicated the event should technically be charged and builders may see a **LOG3**-sized increase.

## PROOF RETENTION

Ordinary-attestation proofs stop verifying after pruning. Under current runtime and archive-node policies, old facts remain re-provable; this is not claimed as immutable.

## ENCODED LEAF

The proven leaf is Attestcoin-specific, not RLP. RPC wrapper fields outside canonical roots, including **from** and **gasUsed**, influence nothing in Cr3dX logic.

## WORKER POLICY

Before each new signature, request a proof again. After signing, recover only by rebroadcasting the exact saved raw transaction.



# The next steps have named conditions, not vague ambition

| DIRECTION            | VALUE                     | WHY DEFERRED                    | UNLOCK CONDITION               |
|----------------------|---------------------------|---------------------------------|--------------------------------|
| Escrow + enforcement | Real recourse             | Needs a write back to source    | Outbound Attestcoin capability |
| Safe reserve release | Stops abandoned locks     | Must block source funding first | Same outbound capability       |
| Sybil resistance     | Identity-level reputation | Outside v0.1 trust boundary     | Issuer identity verification   |
| Worker batching      | Lower per-fact gas        | Atomic-revert recovery cost     | Revisit at real volume         |
| Parallel submission  | Higher throughput         | Reintroduces nonce-lane risk    | Only under measured load       |

Any semantic change invalidates the sealed 63/63 result. Shipping it requires a new independent model freeze and complete replay.

# Every headline number has a transaction, a timestamp or a manifest

**MINED CREDITCOIN GAS**

|                                |         |                   |
|--------------------------------|---------|-------------------|
| Repayment submitAndApply()     | 260,288 | 8 roots           |
| Funding submitAndApply()       | 340,382 | 5 roots           |
| Repayment applyEvidence()      | 292,348 | submitted earlier |
| External race submitAndApply() | 266,056 | 10 roots          |

**SOURCE → FIRST PROOF-READY OBSERVATION → MINED**

|               |             |         |
|---------------|-------------|---------|
| Operation     | Proof-ready | Mined   |
| Repayment     | 1,669.539 s | 1,734 s |
| Funding       | 665.107 s   | 720 s   |
| External race | 588.271 s   | 651 s   |

One testnet run, not an SLA. Exact attestation-transition timestamps were not recorded. Gas rows changed different states and proof shapes; they are evidence records, not a controlled comparison.

**VERIFY IT**

[Live dashboard »](#) [Repository »](#) [Phase B manifests »](#)  
Local: `forge test · npm run test:worker · npm run test:scripts`  
Live: [S5 live runbook »](#) · [S6 worker runbook »](#)